

HW7: Individual Database Project

Option C: Simple Database Application

Company Employee and Project Management System

1. Project Scope

Purpose of the system and intended users

The system manages company departments, employees, projects, and employee project assignments. It allows administrators to maintain employee and project information and generate reports about project workload, salaries, and employee assignments.

Information the database must manage

- Departments (dID, dName)
- Employees (eID, FirstName, LastName, Email, Position, Salary, dID)
- Projects (pID, pName, StartDate, dID)
- WorksOn (eID, pID, Role, HoursWorked)

Important activities represented

- Assigning employees to departments
- Creating projects under departments
- Assigning employees to projects
- Recording employee roles and hours worked
- Searching, updating, and deleting records
- Generating employee, project, and department reports

Part selected from HW6

This project continues from the HW6 Company Employee and Project Management System. The previous design included Staff and Manager as subclasses of Employee. In HW7, these subclass tables were removed. The database was simplified to four core tables: Department, Employee, Project, and WorksOn.

2. Requirements and Business Rules

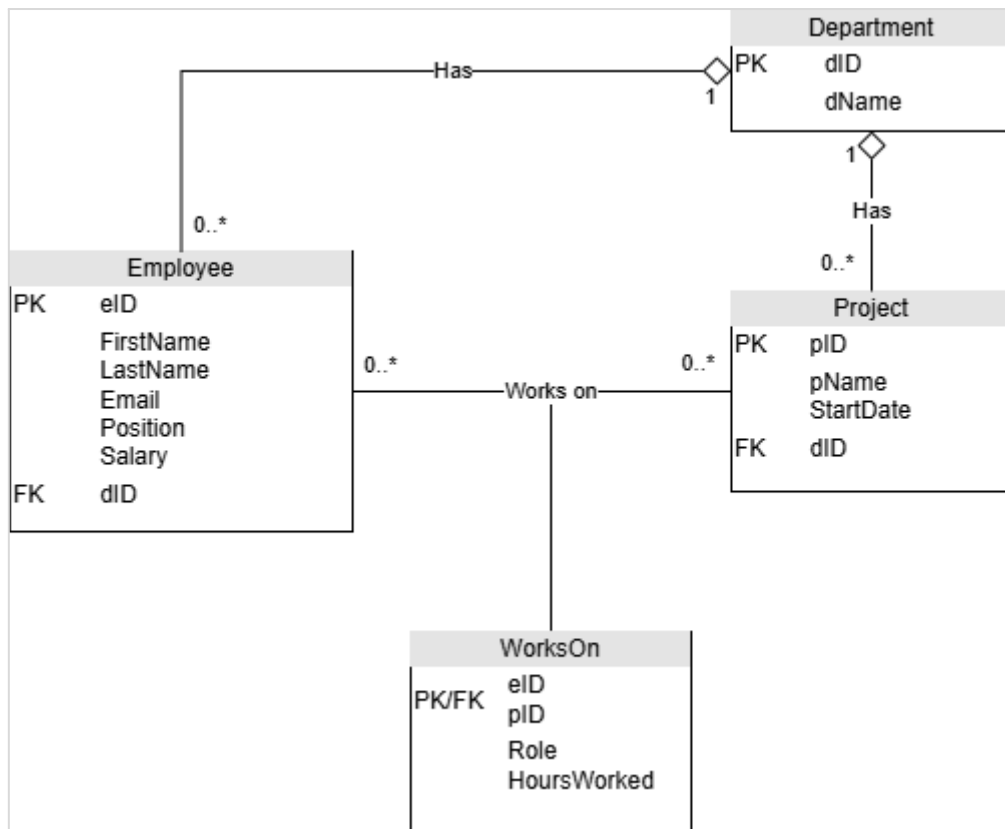
System Requirements

The company needs a database to manage departments, employees, projects, and the employees working on each project. The system must maintain relationships between employees, departments, and projects and record project roles and working hours.

Business Rules

1. Each employee belongs to exactly one department. A department can have zero or many employees.
2. Each project belongs to exactly one department. A department can be responsible for zero or many projects.
3. An employee can work on zero, one, or many projects.
4. A project can have zero, one, or many employees working on it.
5. Each employee has a unique employee ID and email address.
6. Each department has a unique department ID and department name.
7. Each project has a unique project ID and project name.

3. Top-Down Database Design



- The **Department–Employee** relationship is one-to-many because one department can have many employees, while each employee belongs to one department.
- The **Department–Project** relationship is one-to-many because one department can have many projects, while each project belongs to one department.
- The **Employee–Project** relationship is many-to-many through WorksOn.
- **WorksOn** stores Role and HoursWorked because these attributes describe a specific employee's assignment to a specific project.
- **WorksOn** uses (eID, pID) as a composite primary key to prevent the same employee from being assigned to the same project more than once.

4. Functional Dependencies and Keys

Department

dID -> dName

- Candidate key: dID, dName
- Primary key: dID
- dID is short, unique, and can determine each department.

Employee

eID -> FirstName, LastName, Email, Position, Salary, dID

- Candidate key: eID, Email
- Primary key: eID
- Foreign key: dID references Department(dID)
- eID is short, unique, and can determine each employee.

Project

pID -> pName, StartDate, dID

- Candidate key: pID, pName
- Primary key: pID
- Foreign key: dID references Department(dID)
- pID is short, unique, and can determine each project.

WorksOn

(eID, pID) -> Role, HoursWorked

- Candidate key: (eID, pID)
- Primary key: (eID, pID)
- Foreign keys: eID references Employee(eID) and pID references Project(pID)
- Neither eID nor pID alone is unique in WorksOn, so both are required for the primary key.

5. Normalization

Initial Relation

EmployeeProject(eID, FirstName, LastName, Email, Position, Salary, edID, edName, pID, pName, StartDate, pdID, pdName, Role, HoursWorked)

Candidate Key

(eID, pID)

Functional Dependencies

- eID → FirstName, LastName, Email, Position, Salary, edID, edName
- pID → pName, StartDate, pdID, pdName
- (eID, pID) → Role, HoursWorked

First Normal Form (1NF)

All attributes contain atomic values, so the relation satisfies 1NF.

Second Normal Form (2NF)

Partial dependencies on eID and pID are removed by separating employee and project information from the employee-project assignment.

- Employees (eID, FirstName, LastName, Email, Position, Salary, dID, dName)
- Projects (pID, pName, StartDate, dID, dName)
- WorksOn (eID, pID, Role, HoursWorked)

Third Normal Form (3NF)

The dependency dID → dName is separated into the Department table to remove the transitive dependency.

- Departments (dID, dName)

Final Normalized Relations

- Department(dID, dName)
- Employee(eID, FirstName, LastName, Email, Position, Salary, dID)
- Project(pID, pName, StartDate, dID)
- WorksOn(eID, pID, Role, HoursWorked)

Reflection

Yes. The top-down UML design and bottom-up normalization lead to equivalent schemas. The UML design defines the relationships, while normalization removes redundancy. Both approaches result in the same four tables.

6. Database Implementation

```
Active:\Local | hw7_6810545476
1  -- HW7: Option C
2  -- Domain: Company Employee and Project Management
3
4  >Run
5  DROP DATABASE IF EXISTS hw7_6810545476;
6  >Run
7  CREATE DATABASE hw7_6810545476;
8  >Run
9  USE hw7_6810545476;
10
11 -- Create Tables
12
13 >Run | Select
14 CREATE TABLE Department (
15     dID CHAR(4) PRIMARY KEY,
16     dName VARCHAR(100) NOT NULL UNIQUE
17 );
18
19 >Run | Select
20 CREATE TABLE Employee (
21     eID CHAR(4) PRIMARY KEY,
22     FirstName VARCHAR(100) NOT NULL,
23     LastName VARCHAR(100) NOT NULL,
24     Email VARCHAR(100) NOT NULL UNIQUE,
25     Position VARCHAR(100) NOT NULL,
26     Salary DECIMAL(9,2) NOT NULL CHECK (Salary >= 0),
27     dID CHAR(4) NOT NULL,
28     FOREIGN KEY (dID) REFERENCES Department(dID)
29     ON UPDATE CASCADE
30 );
```

```
54 -- Sample Data
55
56 >Run | Select
57 INSERT INTO Department
58 (dID, dName) VALUES
59 ('D001', 'Engineering'),
60 ('D002', 'Human Resources'),
61 ('D003', 'Finance'),
62 ('D004', 'Marketing'),
63 ('D005', 'Information Technology');
64
65 >Run | Select
66 INSERT INTO Employee
67 (eID, FirstName, LastName, Email, Position, Salary, dID) VALUES
68 ('E001', 'Alice', 'Johnson', 'alice.johnson@company.com', 'Software Engineer', 85000.00, 'D001'),
69 ('E002', 'Brian', 'Smith', 'brian.smith@company.com', 'HR Specialist', 68000.00, 'D002'),
70 ('E003', 'Clara', 'Williams', 'clara.williams@company.com', 'Finance Manager', 95000.00, 'D003'),
71 ('E004', 'Daniel', 'Brown', 'daniel.brown@company.com', 'Marketing Specialist', 70000.00, 'D004'),
72 ('E005', 'Emma', 'Davis', 'emma.davis@company.com', 'Systems Engineer', 88000.00, 'D005');
73
74 >Run | Select
75 INSERT INTO Project
76 (pID, pName, StartDate, dID) VALUES
77 ('P001', 'Mobile Application Development', '2026-01-15', 'D001'),
78 ('P002', 'Employee Portal', '2026-02-01', 'D002'),
79 ('P003', 'Annual Budget Planning', '2026-01-20', 'D003'),
80 ('P004', 'Website Redesign', '2026-03-10', 'D004'),
81 ('P005', 'Cloud Migration', '2026-02-15', 'D005');
```

7. Practical Database Use

```
C:\Windows\system32\cmd.exe x + v
Company Employee & Project Management System

Available commands: insert / display / search / report / update / delete / reset / clear / quit
Choose a command: display

Available tables:
1. Department
2. Employee
3. Project
4. WorksOn
Choose a table (or 'back'): 4



| eID  | pID  | Role                | HoursWorked |
|------|------|---------------------|-------------|
| E001 | P001 | Lead Developer      | 150         |
| E001 | P005 | Backend Developer   | 60          |
| E002 | P002 | Project Coordinator | 90          |
| E002 | P004 | Content Reviewer    | 40          |
| E003 | P002 | Finance Advisor     | 30          |
| E003 | P003 | Budget Owner        | 80          |
| E004 | P001 | Marketing Liaison   | 35          |
| E004 | P004 | UI Specialist       | 100         |
| E005 | P001 | Systems Engineer    | 45          |
| E005 | P005 | Project Lead        | 120         |



Available tables:
1. Department
2. Employee
3. Project
4. WorksOn
Choose a table (or 'back'): back

Available commands: insert / display / search / report / update / delete / reset / clear / quit
Choose a command:
```

```
C:\Windows\system32\cmd.exe x + v
Company Employee & Project Management System

Available commands: insert / display / search / report / update / delete / reset / clear / quit
Choose a command: search

Available tables:
1. Department
2. Employee
3. Project
4. WorksOn
Choose a table (or 'back'): 2

Available fields:
1. eID
2. FirstName
3. LastName
4. Email
5. Position
6. Salary
7. dID
Choose a field (or 'back'): 2

Enter search value (or 'back'): X

> No records found where FirstName matches 'X'.

Available fields:
1. eID
2. FirstName
3. LastName
4. Email
5. Position
6. Salary
7. dID
Choose a field (or 'back'): back

Available tables:
1. Department
2. Employee
3. Project
4. WorksOn
Choose a table (or 'back'): back

Available commands: insert / display / search / report / update / delete / reset / clear / quit
Choose a command:
```

8. Validation and Testing

Insert

```
C:\Windows\system32\cmd.exe x + v
Company Employee & Project Management System

Available commands: insert / display / search / report / update / delete / reset / clear / quit
Choose a command: insert

Available tables:
1. Department
2. Employee
3. Project
4. WorksOn
Choose a table (or 'back'): 1

Insert into Department



| dID  | dName                  |
|------|------------------------|
| D001 | Engineering            |
| D002 | Human Resources        |
| D003 | Finance                |
| D004 | Marketing              |
| D005 | Information Technology |



Enter dID (or 'back'): D006
Enter dName (or 'back'): AI

> Record inserted successfully.

Insert into Department



| dID  | dName                  |
|------|------------------------|
| D001 | Engineering            |
| D002 | Human Resources        |
| D003 | Finance                |
| D004 | Marketing              |
| D005 | Information Technology |
| D006 | AI                     |



Enter dID (or 'back'): |
```

Update

```
C:\Windows\system32\cmd.exe x + v
Company Employee & Project Management System

Available commands: insert / display / search / report / update / delete / reset / clear / quit
Choose a command: update

Available tables:
1. Department
2. Employee
3. Project
4. WorksOn
Choose a table (or 'back'): 1

Update Department



| dID  | dName                  |
|------|------------------------|
| D001 | Engineering            |
| D002 | Human Resources        |
| D003 | Finance                |
| D004 | Marketing              |
| D005 | Information Technology |
| D006 | AI                     |



Enter dID (or 'back'): D006

Editable fields:
1. dName
Choose a field to update (or 'back'): 1

Current value: AI
Enter new value for dName (or 'back'): Artificial Intelligence

> Record updated successfully.

Update Department

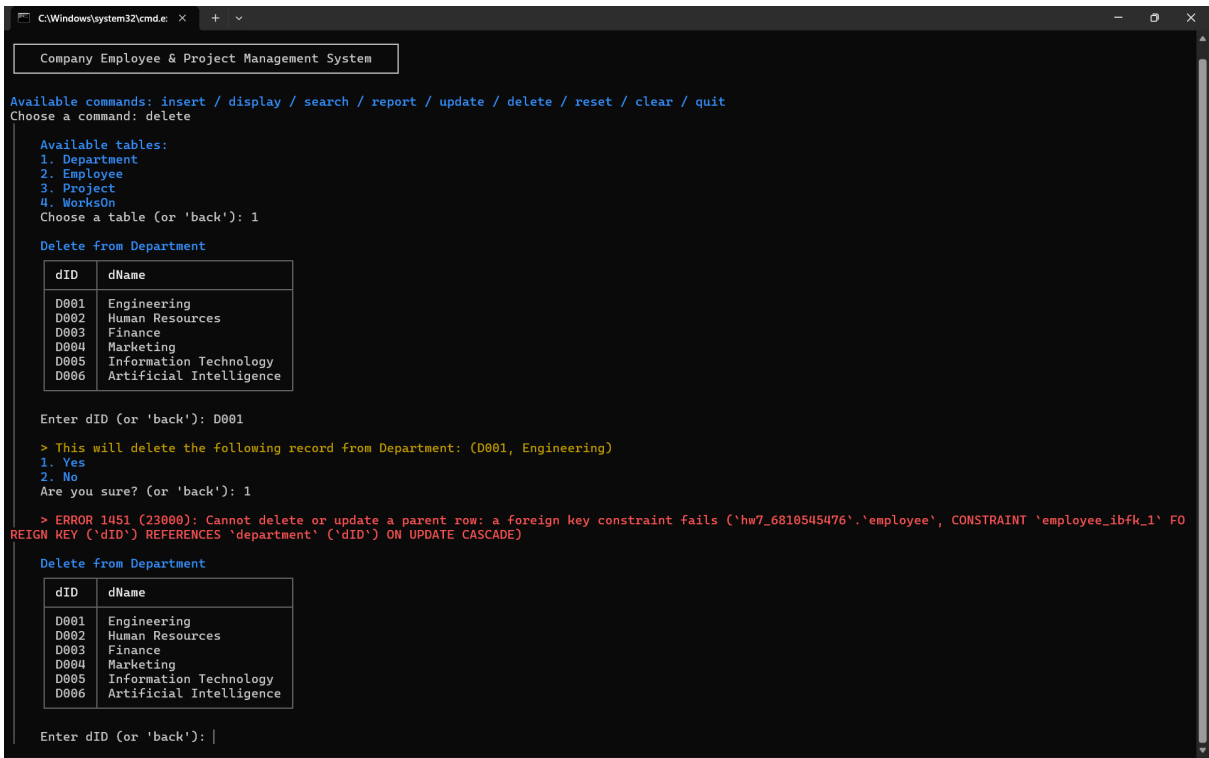


| dID  | dName                   |
|------|-------------------------|
| D001 | Engineering             |
| D002 | Human Resources         |
| D003 | Finance                 |
| D004 | Marketing               |
| D005 | Information Technology  |
| D006 | Artificial Intelligence |

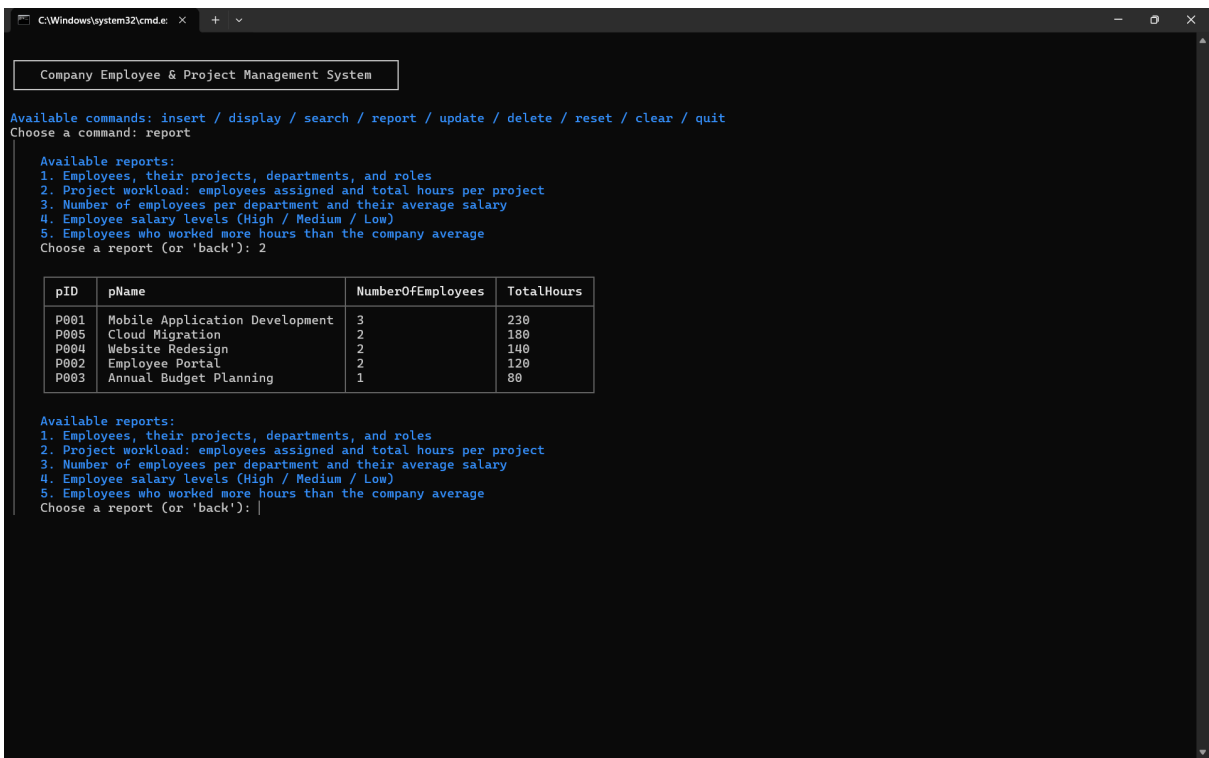


Enter dID (or 'back'): |
```

Constraint



Result involving data from multiple tables / Report



The tests verify that records can be managed correctly, invalid data is rejected by database constraints, and information from multiple tables can be retrieved through SQL joins.